

Lesson 7

Dynamics of continuous systems – bending vibrations of beams

1 Bending vibrations of beams: assumptions

In lesson 2 the problem of axial vibrations of beams was introduced as one of the simplest possible examples for the dynamics of continuous systems. In this lesson, we continue the study of continuous systems, dealing with the bending vibrations of a beam with constant section.

With respect to the subject of the previous lesson, bending vibrations of the beams are slightly more complicate to be treated, due to a higher order of the spatial derivatives appearing in the partial derivatives differential equation describing the motion of the system. On the other hand, bending vibrations are often of greater interest to the engineer, since for a given beam (with specified parameters of section, length, material) they usually occur in a lower frequency range than axial vibrations, which normally makes more important the study of possible dynamic effects like e.g. resonances.

We will treat the problem of the bending vibrations of a beam under the same assumptions that were introduced in the study of axial vibrations, recalled hereafter:

- 1) *Linear elastic constitutive law*: we will consider a linear, isotropic relationship between stress and strain in the beam material; this assumption is normally well satisfied by metallic materials (steel, aluminium) until the plastic yield limit is achieved.
- 2) *Constant section and homogeneous material*: we will consider the section of the beam as constant in shape and dimensions along the beam axis. Furthermore, we will assume that all physical properties of the beam material (density, modulus of elasticity) do not depend upon the particular position inside the beam volume.

Additionally, the following assumptions are made:

- 3) *Plane bending of the beam* is studied, assuming the section of the beam has two axes of symmetry which coincide with the y and z axes of the cartesian reference, and that bending occurs along the z direction, see Section 2 for more details.
- 4) *Bending decoupled from torsion*: assuming the section of the beam has two axes of symmetry also implies that the bending motion is decoupled from the torsional vibrations of the beam.
- 5) *Shear deformation is neglected*, i.e. the “Euler-Bernoulli” theory is used to describe the relationship between bending deformation and the internal forces, see Section 2. Additionally, we assume the effect of rotatory inertia in the beam may be neglected (the consequence of this further assumption is discussed in detail below). We observe that the “Euler-Bernoulli” theory may be considered acceptable for “thin beams”, which means beams whose length L is much greater than the height h of the section (that is the dimension of the section in the plane where bending takes place) so that:

$$\frac{h}{L} \ll 1$$

It is worth recalling that more comprehensive (but also more complex) theories exist for the bending vibrations of a beam, that allow to incorporate the effect of shear deformation and/or of beam rotatory inertia. The use of these theories is recommended in the study of “thick” beams, where the ratio h/L assumes values in the range $\frac{1}{10}$ or more.

2 Bending deformation of a slender beam (Euler-Bernoulli model)

Bending of a beam consists of a displacement of each section in a direction perpendicular to the beam's axis, accompanied by a rotation of the section. We consider the usual choice of a cartesian reference shown in Figure 2.1, with the x axis coinciding with the axis of the beam, the z axis vertical and the y axis forming a right-handed cartesian reference together with x and z . Transversal displacements are then described by the v and w components of displacement. We note there are two distinct components of bending displacement, along the y and z direction, but it can be shown that these two components of bending deformation are decoupled (i.e. bending in one direction does not affect bending in the other direction), provided the y and z axes are symmetry axes of the section. Actually, a weaker condition than the one just stated is enough to conclude that the two bending components are decoupled, but for the sake of simplicity we assume in the following the section of the beam is symmetric with respect to both the y and z axes. This also allows to conclude that bending is decoupled from torsion of the beam. Therefore, from now on, we will confine our attention to the single bending component corresponding to a motion of the beam in the x - z plane, which means the v component of displacement (along axis y) is zero at all point in the beam.

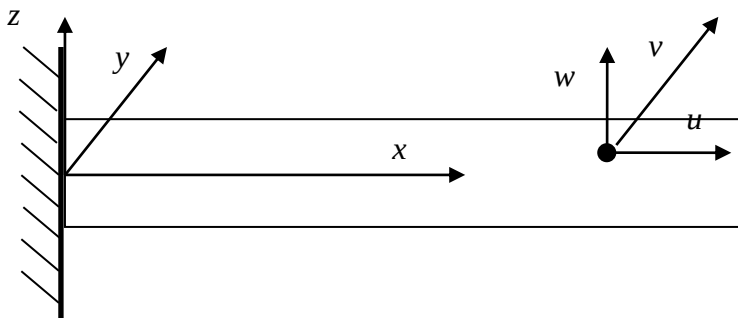


Figure 2.1 Cartesian reference and scalar components of displacements.

Bending is caused by the combined effect of shear forces and bending moments acting as non-zero internal force components in the beam. Additionally, the presence of a static axial internal force may alter the bending stiffness of the beam, see the book by Cheli & Diana “Advanced Dynamics of Mechanical Systems, Section 3.3.1, but this effect is not addressed here for the sake of simplicity.

Assuming small displacements and a linear elastic constitutive law for the material, a linear problem is obtained and hence the principle of super-imposition applies. This means, the total deformable motion caused by the effect of shear forces and bending moments can be seen as the superimposition of the deformation caused by shear forces only and by bending moments only, as shown schematically in Figure 2.2. However, it can be shown that, as long as the beam can be considered slender, see Section 1 above, the deformation caused by shear forces is small compared to the one caused by the bending moments and can be neglected. Hence, a so-called “pure bending” condition is obtained and is studied below.

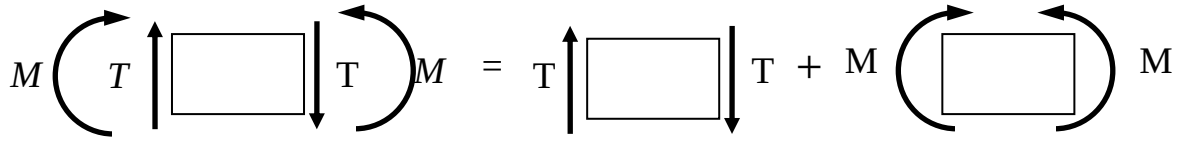


Figure 2.2 Composition of shear forces and bending moments.

The combined effect of shear forces and bending moment produces a transversal displacement and rotation of the section. In pure bending condition, or in case shear deformation caused by shear forces can be neglected due to the slenderness of the beam, the rotation of the section is such that, after the bending deformation has taken place, the section remains orthogonal to the deformed axis of the beam. This allows to define the entire motion of a generic section located at abscissa x by means of a single function $\bar{w}(x)$ representing the transversal displacement of the centre of the section. As shown in Fig. 2.3, due to the transversal displacement of the beam's axis $\bar{w}(x)$, the section rotates by an angle α . In order for the section to be orthogonal to the deformed axis, the rotation α must be equal to the angle formed by the tangent to the deformed beam's axis at x and the undeformed axis of the beam. Assuming the displacement of the axis $\bar{w}(x)$ is small (as usual with the mechanics of flexible systems, we assume the non-rigid motion of the body to be small), this angle can be expressed as:

$$\alpha = \frac{\partial \bar{w}}{\partial x}$$

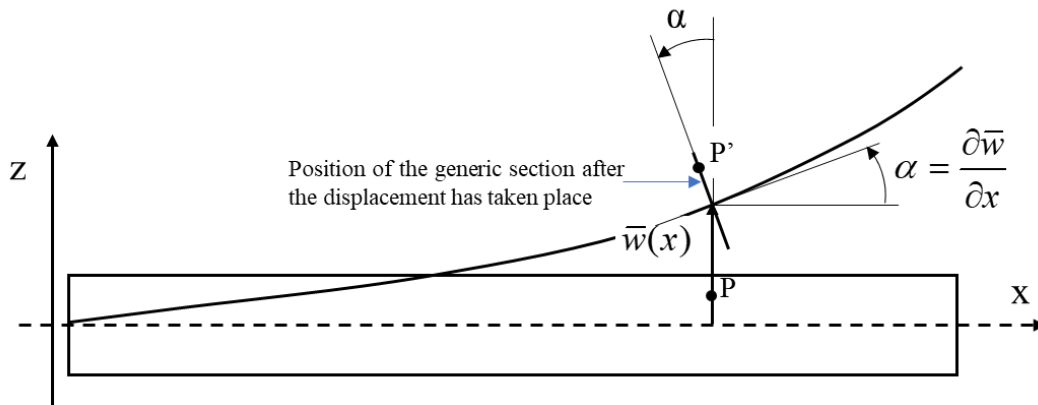


Figure 2.3 Motion of a generic beam section located at abscissa x in pure bending.

We can then describe the motion of a generic point P located at coordinates x, y, z in the undeformed volume of the beam as:

$$\begin{aligned} u &= -z \sin(\alpha) \cong -z\alpha = -z \frac{\partial \bar{w}}{\partial x} \\ v &= 0 \\ w &= \bar{w} - z(1 - \cos(\alpha)) = \bar{w} \end{aligned} \quad (2.1)$$

From the above expressions, the 6 independent components of strain can be obtained as follows:

$$\begin{aligned}
\varepsilon_x &= \frac{\partial u}{\partial x} = -z \frac{\partial^2 \bar{w}}{\partial x^2} ; \quad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = 0 \\
\varepsilon_y &= \frac{\partial v}{\partial y} = 0 ; \quad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} = 0 \\
\varepsilon_z &= \frac{\partial w}{\partial z} = 0 ; \quad \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} = 0
\end{aligned} \tag{2.2}$$

and then, from the linear elastic constitutive law, the normal stress σ_x is obtained:

$$\sigma_x = E \varepsilon_x = -Ez \frac{\partial^2 \bar{w}}{\partial x^2} \tag{2.3}$$

The distribution of stresses corresponding to Eq. (2.3) is shown in Figure 2.4 for a positive value of the curvature of the deformed axis $\frac{\partial^2 \bar{w}}{\partial x^2}$.

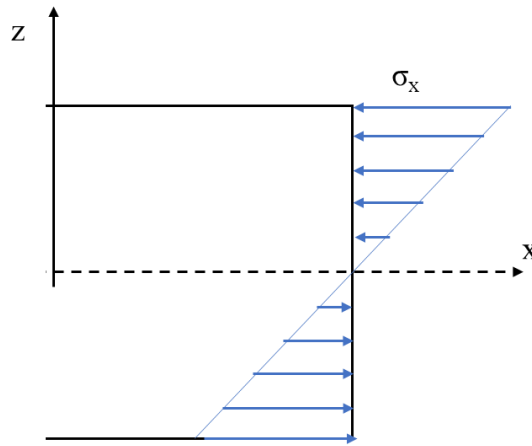


Figure 2.4 Distribution of the normal stresses σ_x along the z direction for a positive curvature of the deformed beam's axis.

The integration of the stress distribution over the surface A of the beam's section provides:

$$\begin{aligned}
N &= \int_A \sigma_x dA = -E \int_A z dA \frac{\partial^2 \bar{w}}{\partial x^2} = 0 \\
T &= \int_A \tau_{xz} dA = 0 \\
M &= \int_A -\sigma_x z dA = E \int_A z^2 dA \frac{\partial^2 \bar{w}}{\partial x^2} = EI_y \frac{\partial^2 \bar{w}}{\partial x^2}
\end{aligned} \tag{2.4}$$

In Eq. (2.4) we note $N=0$ due to the anti-symmetry of the normal stresses in z direction, whereas we assume the section is symmetric with respect to the y axis. We also note $T=0$ because the tangential stresses acting in the section are uniformly equal to zero. Finally, the third line in Eq. (2.4) states that the bending moment M is proportional to the curvature of the deflected axis through a constant which depends on material properties (through the modulus of elasticity E) and on the shape and size of the section through parameter I_y which is called the *area moment of inertia of the section* around axis y . This term is defined as:

$$I_y = \int_A z^2 dA \tag{2.5}$$

In SI units, this parameter is expressed in m^4 and its value depends, as said, by the shape and size of the beam's section. Values of this parameter for typical shapes of the section can be found in handbooks and the formulae for a rectangular and circular shape of the section are reported in Fig. 2.5.

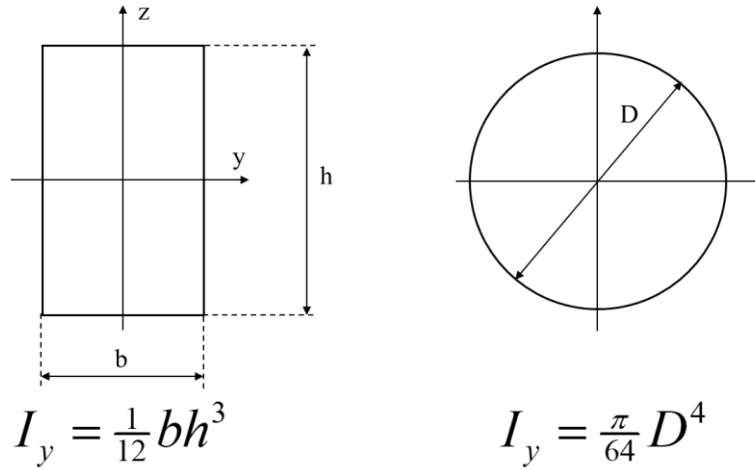


Figure 2.5 Expression of the area moment of inertia I_y for a rectangular and circular section.

From now on, a simpler notation will be used, denoting by $w(x)$ instead of $\bar{w}(x)$ the transversal displacement of the beam's axis along z and dropping subscript “ y ” for the *area moment of inertia* I_y . Hence, the relationship between the deflection of the beam's axis due to bending and the bending moment will be expressed by:

$$M = EI_y \frac{\partial^2 w}{\partial x^2} \quad (2.6)$$

From now on, we skip subscript “ y ” to have a simpler notation, i.e.:

$$M = EI \frac{\partial^2 w}{\partial x^2} \quad (2.6b)$$

3 Equation of motion of bending vibrations for a slender beam

In order to study the bending motion of a flexible beam, we consider the system represented in figure 3.1. We consider the same Cartesian reference already used for the study of axial vibrations, and we confine ourselves to the study of the bending motion in the x - z plane. Finally, we denote by $w(x,t)$ the transversal displacements of the points on the axle of the beam.

When dealing with pure bending of a beams, axial internal forces will be zero everywhere inside the beam, while non-zero shear forces T and bending moments M will take place. Figure 3.1 also shows the sign conventions introduced for the shear forces and bending moments.

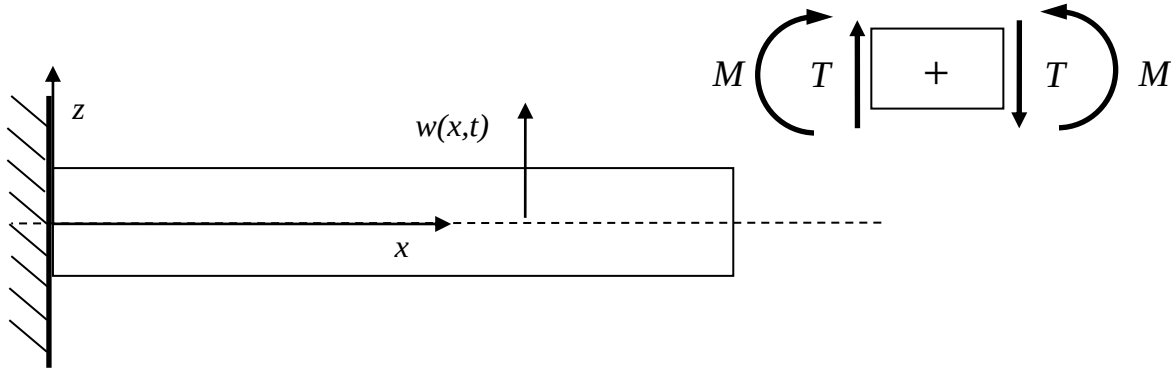


Figure 3.1: Components of internal force and displacement in the study of bending vibrations of beams

In order to write the equation of motion for the bending vibration of the beam, we consider a beam element of infinitesimal length dx , enclosed between the two sections at positions x and $x+dx$, and we introduce the forces acting on the element (see figure 3.2) that are, in absence of external forces:

- the shear force $T(x,t)$ and the bending moment $M(x,t)$ acting on the left face of the element;
- the shear force $T(x+dx,t)$ and the bending moment $M(x+dx,t)$ acting on the right face of the element;
- the inertia force, expressed as the product of the infinitesimal mass of the element dm times the acceleration of the element $\frac{\partial^2 w}{\partial t^2}$, with the sign changed.

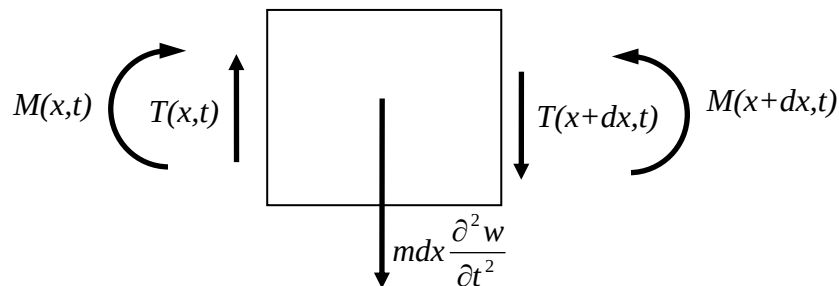


Figure 3.2: Equilibrium of a beam element with infinitesimal dimension dx .

due to the fact that the “Euler-Bernoulli” theory for the beam is used, we may consider as negligible the inertia torque expressed as the product of the beam element moment of inertia times the angular

acceleration of the element with the sign changed (this is what was meant in the list of assumptions as “negligible rotatory inertia”).

Using the D’Alembert principle, we first write the moment equation of motion about an axis perpendicular to the plane of bending (x - z plane, see Figure 3.1) and passing through the centre of the infinitesimal element:

$$-M(x,t) + M(x+dx,t) - T(x,t)\frac{dx}{2} - T(x+dx,t)\frac{dx}{2} = 0$$

then we express the shear force and the bending moment in the “incremented” abscissa $x+dx$ using the differential of the functions bending moment and shear force, so that we get:

$$-M(x,t) + M(x,t) + \frac{\partial M}{\partial x}dx - T(x,t)\frac{dx}{2} - T(x,t)\frac{dx}{2} - \frac{\partial T}{\partial x}\frac{dx^2}{2} = 0 \quad (3.1)$$

finally, we observe that the last term in the above equation may be neglected since it is a higher order infinitesimal, and we rewrite the equation by taking the obvious simplifications and rearrangements as:

$$\left(\frac{\partial M}{\partial x} - T(x,t)\right)dx = 0 \rightarrow T(x,t) = \frac{\partial M}{\partial x} \quad (3.2)$$

where we recognise that the function $T(x,t)$ representing the distribution of shear forces inside the beam is the partial derivative with respect to the space variable x of the bending moment $M(x,t)$.

We write then a second equation for the infinitesimal element in figure 3.2, corresponding to the force equilibrium along the vertical direction:

$$T(x,t) - T(x+dx,t) = mdx \frac{\partial^2 w}{\partial t^2}$$

with the same expansion of the term $T(x+dx,t)$ as in the previous equation we get:

$$T(x,t) - T(x,t) - \frac{\partial T}{\partial x}dx = mdx \frac{\partial^2 w}{\partial t^2} \rightarrow \frac{\partial T}{\partial x} = -m \frac{\partial^2 w}{\partial t^2}$$

and, considering equations (1.2) and (1.1), we get:

$$\frac{\partial T}{\partial x} = \frac{\partial^2 M}{\partial x^2} = EI \frac{\partial^4 w}{\partial x^4}$$

so that the final form of the partial derivatives differential equation describing the motion of the infinitesimal element is:

$$EI \frac{\partial^4 w}{\partial x^4} = -m \frac{\partial^2 w}{\partial t^2} \quad (3.3)$$

we observe that equation (3.3) is formally similar to the equation describing the *axial* vibration of the beam. However, in the case of bending vibrations, the fourth order partial derivatives with respect to the space variable x are involved, instead than the second order derivatives as was the case for axial vibrations. Corresponding to the higher order of derivatives involved, the solution for equation (3.3) is slightly more complicated than for the case of axial vibrations. In particular the study of propagative solutions is more difficult and will not be treated here, also because (as discussed in the previous lesson) stationary solutions are more relevant to engineering problems. The stationary solutions for the bending vibration of a beam are introduced in section 4.

4 Bending vibrations of beams: stationary solutions

As in the case of beam axial vibrations treated in lesson 2, we assume a stationary solution of the bending motion in the form:

$$w(x, t) = \alpha(x)\beta(t) \quad (4.1)$$

where $\alpha(x)$ is a function of space alone describing the waveform of the stationary vibration and $\beta(t)$ is a time dependent vibration amplitude coefficient. To derive appropriate expressions for $\alpha(x)$ and $\beta(t)$ we observe that:

$$\frac{\partial^4 \alpha(x)\beta(t)}{\partial x^4} = \alpha^{IV}(x)\beta(t) \quad ; \quad \frac{\partial^2 \alpha(x)\beta(t)}{\partial t^2} = \alpha(x)\ddot{\beta}(t)$$

where $\alpha^{IV}(x)$ stands for the fourth derivative of α wrt x . By introducing the above relationships in eq. (3.3) we get:

$$EI\alpha^{IV}(x)\beta(t) = -m\alpha(x)\ddot{\beta}(t)$$

According to the “separation of variables” already used in the case of axial vibration we get:

$$\frac{\ddot{\beta}(t)}{\beta(t)} = -\frac{EI}{m} \frac{\alpha^{IV}(x)}{\alpha(x)} = -\omega^2$$

from which we may separate a second-order differential equation in $\beta(t)$ alone and a fourth-order differential equation in $\alpha(x)$ alone:

$$\begin{aligned} \ddot{\beta}(t) + \omega^2 \beta(t) &= 0 \\ \alpha^{IV}(x) - \frac{m\omega^2}{EI} \alpha(x) &= 0 \end{aligned} \quad (4.2)$$

the first equation, as in the case of axial vibrations, provides the solution:

$$\beta(t) = \beta_0 \cos(\omega t + \varphi) \quad (4.3)$$

to solve the second of equations (4.2) we first simplify the notation by defining:

$$\gamma^4 = \frac{m\omega^2}{EI} \quad (4.4)$$

so that the equation becomes:

$$\alpha^{IV}(x) - \gamma^4 \alpha(x) = 0 \quad (4.5)$$

which is a fourth order, linear differential equation with constant parameters. The solution of this type of equation is a linear combination of four terms having the exponential form:

$$\alpha(x) = \alpha_0 e^{\lambda x} \quad \rightarrow \quad \alpha^{IV}(x) = \lambda^4 \alpha_0 e^{\lambda x}$$

with the constant coefficient λ in the exponent being in general complex valued. By substitution of the above solution in equation (4.5) we get:

$$(\lambda^4 - \gamma^4) \alpha_0 e^{\lambda x} = 0 \quad \rightarrow \quad (\lambda^4 - \gamma^4) \alpha_0 = 0$$

and, to avoid the “trivial” solution $\alpha_0=0$:

$$\lambda^4 - \gamma^4 = 0$$

that provides as solutions the following exponents λ :

$$\lambda_{1,2} = \pm i\gamma$$

$$\lambda_{3,4} = \pm \gamma$$

The solution of eq. (4.5) takes then the complex form:

$$\alpha(x) = \alpha_1 e^{-i\gamma x} + \alpha_2 e^{i\gamma x} + \alpha_3 e^{-\gamma x} + \alpha_4 e^{\gamma x} \quad (4.6)$$

To ensure the solution (4.6) to be real-valued, we replace:

$$\alpha_1 = \frac{A + iB}{2} ; \alpha_2 = \frac{A - iB}{2} ; \alpha_3 = \frac{C - D}{2} ; \alpha_4 = \frac{C + D}{2}$$

with A, B, C, D four real-valued arbitrary constants. Recalling that:

$$\begin{aligned} \cos(\gamma x) &= \frac{e^{i\gamma x} + e^{-i\gamma x}}{2} ; & \sin(\gamma x) &= \frac{e^{i\gamma x} - e^{-i\gamma x}}{2i} \\ \cosh(\gamma x) &= \frac{e^{\gamma x} + e^{-\gamma x}}{2} ; & \sinh(\gamma x) &= \frac{e^{\gamma x} - e^{-\gamma x}}{2} \end{aligned}$$

we obtain a new form for equation (4.6) as follows:

$$\alpha(x) = A \cos(\gamma x) + B \sin(\gamma x) + C \cosh(\gamma x) + D \sinh(\gamma x)$$

and then:

$$w(x, t) = \alpha(x) \beta(t) = [A \cos(\gamma x) + B \sin(\gamma x) + C \cosh(\gamma x) + D \sinh(\gamma x)] \cos(\omega t + \varphi) \quad (4.7)$$

where the amplitude of the time dependent part of the solution (4.3) has been embedded in constants A, B, C, D. Expression (4.7) represents the general solution for the stationary free bending vibration for a constant section Euler-Bernoulli beam, regardless the boundary conditions acting on the beam at its ends. Different values of natural frequencies and modes of vibration are then associated to different boundary conditions at the ends of the beam.

In the next three subsections this point is dealt with: different kinds of possible boundary conditions are first examined and translated into mathematical conditions (section 4.1), then the “pinned-pinned” boundary conditions case is examined in section 4.2, and finally a general procedure to treat the problem for arbitrary boundary conditions and for systems of beams is introduced in section 4.3.

4.1 Different kinds of boundary conditions for the bending motion of a beam

We have seen in lesson 2 that in the problem of beam axial vibrations one boundary condition was written for each end of the beam, thus providing a total of two boundary conditions. It could be demonstrated that the total number of boundary conditions in the problem is always equal to the maximum order of spatial derivatives in the partial derivatives differential equation to be solved. Thus we expect in the case of bending vibrations four boundary conditions, which means in turn two per each end of the beam.

In this section, we will consider three possible cases of boundary conditions that are those most frequently occurring in engineering practice. Other boundary conditions may be introduced (see Diana & Cheli) and their treatment may be deduced from the examples reported hereafter.

The first case we consider is the clamped beam end: this constraint prevents the end of the beam from undergoing any vertical displacement or rotation¹. If we denote by $\bar{x}$ the position of the

¹ We note by the way that it is not relevant to the problem treated here whether or not the constraint is also preventing the horizontal displacement of the section, since axial movements are decoupled from the bending motion analysed.

section where the clamp is placed (this will be either $\bar{x}=0$ for the left end or $\bar{x}=\ell$ for the right end of the beam), we must then require:

$$\begin{aligned} w(x,t)|_{x=\bar{x}} &= 0 \quad \text{for any time } t \\ \frac{\partial w}{\partial x}|_{x=\bar{x}} &= 0 \quad \text{for any time } t \end{aligned} \quad (4.8a)$$

where the second boundary condition is justified by the fact that, assuming shear deformation to be negligible (in accordance with the Euler-Bernoulli beam theory), the rotation of the section coincides with the inclination of the beam axis, which in turn for small displacements is expressed by the derivative of the transversal displacement $w(x,t)$ with respect to the spatial derivative x .

In the case of a clamped end, the two boundary conditions both reflect the presence of a constraint and are then called “geometric” boundary conditions.

As a second case, we consider the “free end” boundary conditions, where the beam is not subjected to any constraint. Obviously in this case it is not possible to write geometric boundary conditions. However, if we consider an infinitesimal element of the beam that includes the free end of the beam, then two equilibrium equations can be written providing two boundary conditions called “natural” boundary conditions. In order to clarify this point we consider (without loss of generality) the case where the left end of the beam is free. The situation is depicted in figure 4.1 below: we may then write the force equation of motion along the vertical direction and the moment equation of motion for the same element:

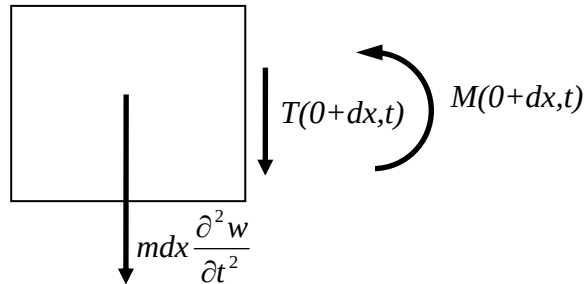


Figure 4.1: Equilibrium of an infinitesimal beam element on the left end of the beam in the case of “free end” boundary condition.

$$T(0,t) + \frac{\partial T}{\partial x} dx + mdx \frac{\partial^2 w}{\partial t^2} = 0$$

$$M(0,t) + \frac{\partial M}{\partial x} dx = 0$$

in both equations, the terms containing the infinitesimal length dx of the element are negligible with respect to the finite terms, so that we may simplify the equations in:

$$T(0,t) = 0$$

$$M(0,t) = 0$$

and, recalling equations (2.6) and (3.2):

$$EI \frac{\partial^3 w}{\partial x^3} \bigg|_{x=0} = 0 \rightarrow \frac{\partial^3 w}{\partial x^3} \bigg|_{x=0} = 0$$

$$EI \frac{\partial^2 w}{\partial x^2} \bigg|_{x=0} = 0 \rightarrow \frac{\partial^2 w}{\partial x^2} \bigg|_{x=0} = 0$$

if we now remove the assumption of considering the left end of the beam, we get the general expression for the free end boundary conditions in a generic extremity of the beam $\bar{x}$:

$$\frac{\partial^3 w}{\partial x^3} \bigg|_{x=\bar{x}} = 0$$

$$\frac{\partial^2 w}{\partial x^2} \bigg|_{x=\bar{x}} = 0 \quad (4.8b)$$

we note by the way that the “natural” boundary conditions introduced in the free end involve higher order derivatives of the solution $w(x,t)$ than the “geometric” boundary conditions. Also we note that in presence of a clamp constraint, it is impossible to write any natural boundary condition, since each of the two equilibrium equations written above in the case of the free end would introduce in that case an unknown force component (either the clamping shear force or the clamping moment) associated with the presence of the constraint.

Finally, we consider a case that may be considered as intermediate between the “clamped end” and the “free end” cases, that is the case of a “pinned end”, which means that the considered beam section is supported by a constraint like a hinge, that prevents the section from any vertical displacement but leaves it free to rotate around a fixed point. We may introduce in this case one single geometric boundary condition, that reflects the constraint acting on the vertical displacement and will thus correspond to the first of conditions (4.8a) found in the case of a clamped section, and additionally one single natural boundary condition, that reflects the possibility for the beam section to freely rotate, and will then correspond to the second of conditions (4.8b), found in the case of a free section. Thus, the two boundary conditions corresponding to a pinned section are:

$$w(\bar{x}, t) = 0$$

$$\frac{\partial^2 w}{\partial x^2} \bigg|_{x=\bar{x}} = 0 \quad (4.8c)$$

4.2 Natural frequencies and modes of vibrations for the pinned-pinned beam

We consider first the case of a pinned-pinned beam to find the associated natural frequencies and modes of vibration. The advantage of considering this case is that the resulting equation is very simple, and allows for an analytical treatment. This is not the case for generic boundary conditions, where the use of a numerical procedure is normally needed, as clarified in the next section. Thus the case of the pinned-pinned beam, besides being particularly interesting from the engineering point of view, also allows to introduce the problem of computing the beam natural frequencies and modes of vibration with the simplest possible mathematical treatment.

If we consider a pinned boundary condition applied at both beam ends $\bar{x}=0$ and $\bar{x}=L$ we get as the total set of four boundary conditions:

$$\begin{aligned}
w(0,t) &= 0 \\
\left. \frac{\partial^2 w}{\partial x^2} \right|_{x=0} &= 0 \\
w(L,t) &= 0 \\
\left. \frac{\partial^2 w}{\partial x^2} \right|_{x=L} &= 0
\end{aligned} \tag{4.9}$$

we replace the general solution (4.7) into (4.9), considering that:

$$\frac{\partial^2 w}{\partial x^2} = \gamma^2 [-A \cos(\gamma x) - B \sin(\gamma x) + C \cosh(\gamma x) + D \sinh(\gamma x)] \cos(\omega t + \psi)$$

so we get from the first and second of (4.9):

$$\begin{cases} A + C = 0 \\ -A + C = 0 \end{cases} \rightarrow \begin{cases} A = 0 \\ C = 0 \end{cases} \tag{4.10}$$

if we now consider the third and fourth of conditions (4.9), taking into account the above result, we get:

$$\begin{cases} B \sin(\gamma L) + D \sinh(\gamma L) = 0 \\ -B \sin(\gamma L) + D \sinh(\gamma L) = 0 \end{cases} \rightarrow \begin{cases} D \sinh(\gamma L) = 0 \\ B \sin(\gamma L) = 0 \end{cases}$$

now, since the hyperbolic sine function is zero only for zero argument and since $\gamma L \neq 0$, the first of the two above conditions necessarily implies:

$$D = 0 \tag{4.11}$$

on the other hand, if we consider the second of the above conditions, we may come out with two possibilities: the first one is $B=0$ that (given $A=C=D=0$) represents the trivial case, with the beam being still in the equilibrium configuration. In order to avoid this trivial solution we set:

$$\sin(\gamma L) = 0 \rightarrow \gamma L = k\pi$$

and finally, recalling (4.4):

$$\frac{m\omega^2}{EJ} = \gamma^4 = \left(\frac{k\pi}{L} \right)^4$$

and solving the above condition for the possible values of the pulsation ω :

$$\omega_k = \left(\frac{k\pi}{L} \right)^2 \sqrt{\frac{EJ}{m}} \quad k = 1, 2, \dots \tag{4.12}$$

where ω_k takes the meaning of the k -th *natural frequency* of the pinned-pinned beam in its bending vibrations. In (4.12) coefficient k may assume any integer positive value, without any upper limit.

In order to define the corresponding k -th mode of vibration we introduce (4.10), (4.11) and (4.12) into the general expression of stationary bending vibrations (4.7) and we get:

$$w_k(x,t) = B_k \sin\left(\frac{k\pi}{L}x\right) \cos(\omega_k t + \varphi_k) \tag{4.13}$$

which shows that the k -th mode of vibration (space function describing the shape of deformation associated with the k -th natural frequency) is the same as for the axial vibrations of the beam, that is a sinusoid having as its wavelength an integer sub-multiple of twice the beam span:

$$\alpha_k(x) = B_k \sin\left(\frac{k\pi}{L}x\right) \quad (4.14)$$

figure 4.2 reports the shape of the first four modes of vibration, and perfectly corresponds to figure 4.2 in the lesson concerning axial vibrations.

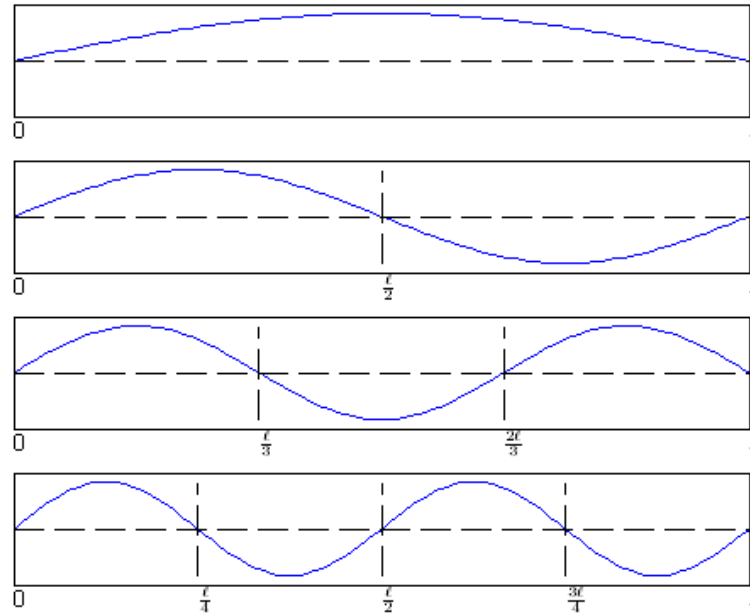


Figure 4.2: Shape of the first four modes of vibration for the bending vibration of a pinned- pinned beam with constant section

The more general bending motion for the pinned-pinned beam in bending motion is then the combination of all motion components of the type (4.13):

$$w(x,t) = \sum_{k=1}^{\infty} w_k(x,t) = \sum_{k=1}^{\infty} B_k \sin\left(\frac{k\pi}{L}x\right) \cos(\omega_k t + \varphi_k) \quad (4.15)$$

where the amplitude parameters B_k and the phase parameters φ_k have to be determined by considering a particular set of initial conditions for the system.

4.3 Natural frequencies and modes of vibrations for other boundary conditions

As explained above, the natural frequencies and modes of vibration of a beam in bending do not only depend upon the physical parameters of the beam (length, section, material), but also upon the particular boundary conditions considered. The same approach described in section 4.2 may be used to define the natural frequencies and modes of vibration for any combination of boundary conditions, with the only difference that, in general, an analytical solution will not be possible.

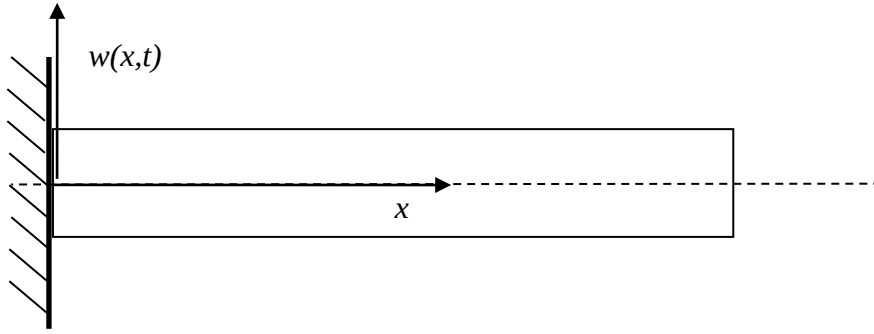


Figure 4.3: The cantilever beam

We provide an example of the general procedure by considering the cantilever beam, clamped at one end and free at the other. In this case, the four conditions to be processed are the two of the clamped end in $x=0$ and the two of the free end in $\bar{x} = L$

$$\begin{aligned} w(0,t) &= 0 \\ \left. \frac{\partial w}{\partial x} \right|_{x=0} &= 0 \\ \left. \frac{\partial^2 w}{\partial x^2} \right|_{x=L} &= 0 \\ \left. \frac{\partial^3 w}{\partial x^3} \right|_{x=L} &= 0 \end{aligned}$$

It is left to the reader to verify that introducing these four conditions in the general solution (4.7) for the stationary bending vibration of the system, the following system of four equations in the unknowns A, B, C, D is obtained:

$$\begin{aligned} A + C &= 0 \\ B + D &= 0 \\ -A \cos(\gamma L) - B \sin(\gamma L) + C \operatorname{Chs}(\gamma L) + D \operatorname{Sh}(\gamma L) &= 0 \\ A \sin(\gamma L) - B \cos(\gamma L) + C \operatorname{Sh}(\gamma L) + D \operatorname{Ch}(\gamma L) &= 0 \end{aligned}$$

which may be rewritten in the form of the following matrix equation:

$$[H(\gamma)] \underline{x} = \underline{0} \quad (4.16)$$

where $\underline{x}$ is the vector formed by the four unknowns:

$$\underline{x} = \begin{Bmatrix} A \\ B \\ C \\ D \end{Bmatrix}$$

and $[H(Y)]$ has the following expression:

$$[H(\gamma)] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -\cos(\gamma L) & -\sin(\gamma L) & \operatorname{Ch}(\gamma L) & \operatorname{Sh}(\gamma L) \\ \sin(\gamma L) & -\cos(\gamma L) & \operatorname{Sh}(\gamma L) & \operatorname{Ch}(\gamma L) \end{bmatrix}$$

By considering any other boundary conditions different from those of the cantilever beam, the same equation (4.16) would have been obtained, but with a different expression for matrix $[H(\gamma)]$. Now, we may observe that in equation (4.16)

- the term at right hand is zero (i.e., the problem is *homogeneous*);
- matrix $[H(\gamma)]$ depends upon parameter γ .

In order to get non-trivial solutions, we must require:

$$\det([H(\gamma)]) = 0 \quad (4.17)$$

which results into a non-linear equation in parameter γ , having an infinite number of solutions γ_k , $k=1,2,\dots$. Using equation (4.4) it is then possible to obtain the corresponding values for the natural frequencies, that will be:

$$\omega_k = (\gamma_k)^2 \sqrt{\frac{EJ}{m}}$$

Finally, by replacing $\gamma = \gamma_k$ into equation (4.16), a solution $\underline{x}_k$ for that equation will be found, being formed by the four parameters A_k, B_k, C_k, D_k , with one of these coefficients set to an arbitrary value. By replacing these four values into solution (4.7), the k -th component of the free motion for the system is defined, with the spatial part of the solution α_k representing the k -th mode of vibration, associated with the k -th natural frequency ω_k .

As we may see, the solution scheme to be adopted in the case of generic boundary conditions is the same as the one used to treat the pinned-pinned case, except for the fact that for generic boundary conditions an analytical treatment of equation (24.16) is not possible, whereas for the pinned-pinned case equation (4.17) is easily transformed into the particular form:

$$\sin(\gamma L) = 0$$

so that the auto-solutions can be determined analytically, as described in section 4.2.

4.4 Natural frequencies and modes of vibrations systems of beams

The solutions for the free motion of beams in bending discussed in sections (4.2) and (4.3) all make reference to the case of a single beam with constant section, which is not subjected to any concentrated force (as could be caused by the presence of a constraint, or a concentrated spring / mass, etc.) except at its ends. However, we easily recognise that the presence of “discontinuities” along the beam axis is often encountered in the engineering practice where variations of the section, constraints and concentrated masses / springs may take place due to the particular design of the system to be studied. This could be the case e.g. of the shaft of a steam turbine + generator, normally resting on several supports that may be represented in first approximation as pins², with a mechanical scheme corresponding to that of figure 4.4

² Typically, the high-pressure, intermediate-pressure, low-pressure stages of the turbine and the generator may be seen as four pinned-pinned beam systems connected each other at their ends trough clamps that are physically realised by flanges, nuts and screws.

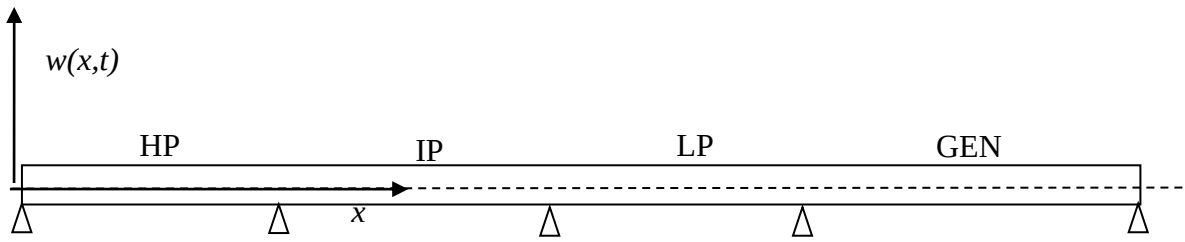


Figure 4.4: Example of a beam resting on multiple supports (steam turbine + generator)

The scheme in figure 4.4 might be also used to represent the situation of a railway / road viaduct, that may also be considered in some cases as a continuous beam resting on multiple supports. We recognise that the situation of figure 4.4 may be described as that of a “system of beams” interconnected each other, where in the specific case each beam is represented by the single span separating two subsequent supports. Aim of this section is to describe how the solution for the free bending vibration of such systems may be computed using the general solution for the free bending vibration of a single beam (4.7).

To this end, we simplify the example in figure 4.4 to the case of a continuous beam over three supports, as represented in figure 4.5: this will allow us to illustrate the procedure in a simple case, but from the treatment it should be clear that the same approach may be used for more complicate systems of beams.

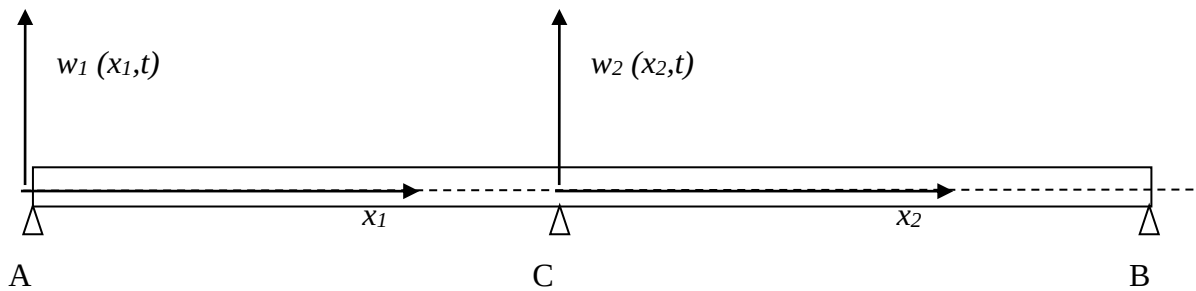


Figure 4.5: Example of a beam resting on three supports

In order to treat the case of figure 4.5, we observe that the assumptions under which we found solution (4.7) are not applicable to the whole beam (due to the presence of the central support), but instead are satisfied if we consider separately the left and right halves (spans) of the beam. Thus, we may introduce two axial coordinates x_1 and x_2 , spanning respectively the left and right halves of the system, and correspondingly two solutions of the type (4.7) that separately apply to the left and right spans of the system:

$$\begin{aligned} w_1(x_1, t) &= [A_1 \cos(\gamma x_1) + B_1 \sin(\gamma x_1) + C_1 \text{Ch}(\gamma x_1) + D_1 \text{Sh}(\gamma x_1)] \cos(\omega t + \varphi) \\ w_2(x_2, t) &= [A_2 \cos(\gamma x_2) + B_2 \sin(\gamma x_2) + C_2 \text{Ch}(\gamma x_2) + D_2 \text{Sh}(\gamma x_2)] \cos(\omega t + \varphi) \end{aligned} \quad (4.17)$$

We observe by the way the time-dependent part of the two solutions must obviously be the same, since the motion of the two halves of the system must be synchronised. From eq. (4.4) it then follows that since parameter ω is equal for the two solutions in (4.17), then also the same γ coefficient has to be used for the two subsystems, as far as the EJ and m coefficients are the same along the whole beam.

To find the natural frequencies and modes of vibration for the system in figure 4.5, we must define a set of appropriate boundary conditions for equation (4.17), which will be referred to both the “external boundaries” (the ends of the system, sections A and B) and the “internal boundary” (the central pin, section C, where the two halves are connected).

For sections A and C, the already encountered boundary conditions of the pinned end take place:

$$\begin{aligned} w_1(0, t) &= 0 \\ \frac{\partial^2 w_1}{\partial x_1^2} \Big|_{x_1=0} &= 0 \\ w_2(L, t) &= 0 \\ \frac{\partial^2 w_2}{\partial x_2^2} \Big|_{x_2=L} &= 0 \end{aligned}$$

Moreover, in point C four additional boundary conditions hold, the first three being geometrical (equality of displacements and rotations at the connection between the two halves of the beam, plus zero displacement due to the presence of the pin) and the fourth one being a natural boundary condition representing the rotational dynamic equilibrium of the infinitesimal beam element cut just before and just after the central pin³:

$$\begin{aligned} w_1(L, t) &= w_2(0, t) \\ \frac{\partial w_1}{\partial x_1} \Big|_{x_1=L} &= \frac{\partial w_2}{\partial x_2} \Big|_{x_2=0} \\ w_1(L, t) &= 0 \\ \frac{\partial^2 w_1}{\partial x_1^2} \Big|_{x_1=L} &= \frac{\partial^2 w_2}{\partial x_2^2} \Big|_{x_2=0} \end{aligned}$$

we note by the way that the total number of boundary conditions (eight) equals the number of unknowns in eq. (4.17): $A_1, B_1, \dots, D_1$ plus $A_2, B_2, \dots, D_2$. The above listed eight boundary conditions may be written as a single matrix equation with the form (4.16), the only difference with the case of the cantilever beam previously treated being that in this case matrix $[H(\gamma)]$ will have dimension 8 instead than 4. Then, the same procedure to find the natural frequency and modes of vibrations described in the end of section 4.3 (based on computing the values of parameter γ that correspond to zero values of the determinant of matrix $[H]$) may be used also for the system of beams in figure 4.5 and, more generally, for any system of beams where all elementary beams have a common axis (like the shaft in figure 4.4).

For systems of beams not satisfying the condition of sharing a common axis, like the one shown in figure 4.6, an even more complicate procedure must be set in place, because axial and bending vibrations of the system are coupled: for instance, bending in the transverse beam (horizontal) is coupled with the axial vibration of the two vertical legs and vice-versa.

As we see then, a procedure may be set to study the free vibration of systems of beams, but this procedure becomes very complicate and difficult to manage as soon as the complexity of the system increases (number of beam parts, relative direction of the beam axis etc.). Moreover, the procedure illustrated in this section is not general: any different system of beams will require a specific treatment, with the number of solutions in the form (4.7) and correspondingly the number and type of boundary conditions depending upon the particular case treated.

Thus, we understand the need for a more flexible and general approach, allowing to treat in the same way a wide category of continuous systems and possibly allowing for a codification of the solution algorithm into a software code for the computer-based solution of the problem. These

³ The vertical equilibrium of the same element cannot be written as a boundary condition, due to the presence of the unknown vertical reaction force introduced by the pin

properties and advantages are actually provided by the finite element approach, that will be treated later in the course

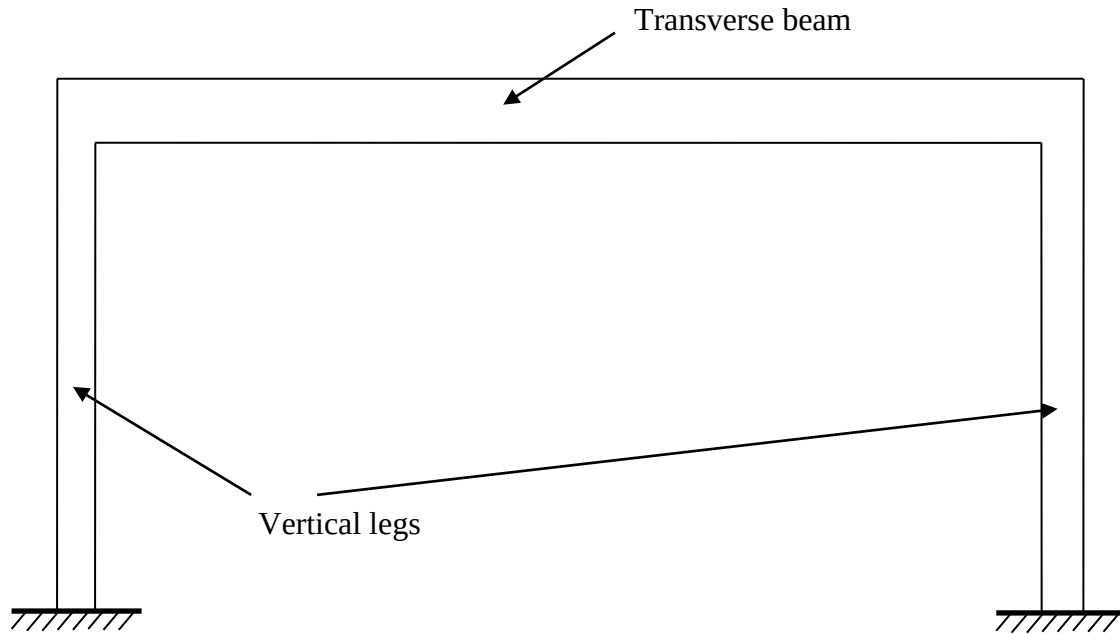


Figure 4.6: Example of a system where the different beam parts are not aligned along a common axis; for this type of system bending and axial vibrations are coupled each other.